

[illegible]

- Use a USB-C **data** cable, not a charging-only cable.
- Install Arduino IDE from **[arduino.cc/en/software](https://www.arduino.cc/en/software)**.
- No separate LED library is needed; megaTinyCore includes it.

- 1 In Arduino IDE, open **File** → **Preferences** (Arduino IDE → Settings on macOS).
- 2 Add this to **Additional Boards Manager URLs**:

3 Open **Tools** → **Board** → **Boards Manager**, search for **megaTinyCore**, and install the newest version by Spence Konde.

Plug the board into the computer. The power LED should turn on and a new serial port should appear.

## Cursed PCB · Quick Start

# Make some light

## Get and upload the example

### Code and full instructions

[github.com/user-will/cursedPCB](https://github.com/user-will/cursedPCB)

- 1 Download the repository and open `examples/LED_example/LED_example.ino`.
- 2 Choose **Sketch** → **Verify/Compile**.
- 3 Choose **Sketch** → **Upload Using Programmer**. Do not use the regular Upload command.

The LEDs show red, green, blue, white, a rainbow sweep, and a moving white chase.

## Optional mode button

Connect a normally-open pushbutton between header pins **0 (PA6)** and **2 (PA1)**. No external resistor is needed with the example sketch. Each press advances the animation. The LEDs still work without the button.

## Six-pin header

Component side up, USB-C at the bottom; read left to right:

| Label | ATTiny | Use                |
|-------|--------|--------------------|
| 5V    | VDD    | Regulated 5 V rail |
| GND   | GND    | Ground             |
| 0     | PA6    | Digital, DAC, TX   |
| 1     | PA7    | Digital, RX        |
| 2     | PA1    | Digital, I²C SDA   |
| 3     | PA2    | Digital, I²C SCL   |

The LEDs use **PA3 / Arduino pin 4**. PA0 is reserved for UPDI programming. The onboard USB connection programs UPDI; it is not a Serial Monitor connection.

## Troubleshooting

- **No port:** try a known data cable or install a CH340 driver.
- **Upload error:** recheck board, BOD voltage, and programmer; close Serial Monitor.
- **Still failing:** select *SerialUPDI - SLOW: 57600 baud*.
- **Dark LEDs:** use `PIN_PA3`, four pixels, and `NEO_GRB`.

## Power safely

Use USB-C for normal power. Never apply more than regulated 5 V to the exposed 5V pin. Do not connect external power while USB-C is connected. Disconnect power before changing wiring. Avoid shorts and staring into LEDs at high brightness.